

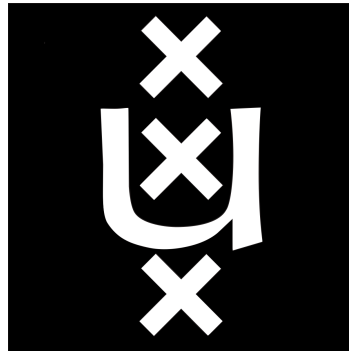
AI health diagnosis chatbots, a multimodal approach

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Key words

Health diagnosis

Health diagnosis is the process of determining a patient's condition by integrating information from their history, physical examination, and diagnostic tests. It relies on probabilistic reasoning and pattern recognition under uncertainty. In computational settings, diagnostic systems learn mappings from clinical inputs to probabilistic disease labels.

Chatbot

A chatbot is a computer program or artificial intelligence system designed to simulate conversation with human users, through auditory or textual methods (Shawar & Atwell, 2007; Adamopoulou & Moussiades, 2020). Chatbots use natural language processing techniques to interpret user inputs and generate contextually appropriate responses (Jurafsky & Martin, 2023; Perez & Pascual, 2018).

AI Chatbot

An AI chatbot uses artificial intelligence techniques such as natural language processing, deep learning, and reinforcement learning to interpret user intent and generate dynamic, context-aware responses (Jurafsky & Martin, 2023; Perez & Pascual, 2018). AI chatbots can improve their performance over time through exposure to conversational data and are capable of handling a larger range of user inputs with higher flexibility and relevance (Adamopoulou & Moussiades, 2020).

Bias

Bias in artificial intelligence and machine learning refers to systematic and unfair deviations in outcomes or predictions caused by flawed assumptions, imbalanced data, or structural inequalities embedded in training data or algorithms (Barocas et al., 2019; Mehrabi et al., 2021).

Multimodal Approach

A multimodal approach refers to the integration and joint analysis of information from multiple data types, such as text, speech, images, video, and sensor data to improve the performance and robustness of artificial intelligence systems (Baltrušaitis et al., 2019; Kiela & Bottou, 2014).

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1. Introduction

Artificial intelligence has been having an increasing positive impact on the healthcare industry. It does this by enabling systems that can assist in diagnosis, triage and patient communication. In this context, AI powered health chatbots have seen a significant rise. They are systems designed to simulate conversation with human users, through auditory or textual methods (Shawar & Atwell, 2007). Chatbots make use of natural language processing and machine learning to interpret inputs and generate responses. Unlike rule-based chatbots, AI chatbots can improve their performance over time through empirical conversational data. They can also handle a bigger range of inputs with higher flexibility and relevance (Adamopoulou & Moussiades, 2020). These systems have a high potential to assist healthcare systems and physician availability (Radziwill and Benton, 2017).

Many current chatbots are based on unimodal language models that only process textual input. And even though these systems have achieved considerable success, they do not incorporate other important and high-potential modalities, such as medical images or audio signals (Perez and Pascual, 2018; Jurafsky and Martin, 2023). The goal of multimodal machine learning is to overcome these limitations by allowing models to process multiple data modalities. By using complementary strengths of different data types, multimodal systems are better at interpreting complex phenomena. They thus improve in tasks such as emotion recognition, visual question answering, and human-computer interaction (Ngiam et al., 2011; Liang et al., 2022). In the medical field, this opens possibilities such as combining radiology images with clinical notes for better diagnosis (Baltrušaitis, Ahuja and Morency, 2019; Liang et al., 2022). Yet while multimodal systems are promising, they also raise questions of bias, and interpretability, which are critical issues in medical decision-making (Barocas, Hardt and Narayanan, 2019; Mitchell et al., 2021).

In this paper, we compare unimodal (text only) and multimodal (text and image) health diagnosis chatbots, focusing on two key outcomes: diagnostic accuracy across diverse medical tasks, and demographic bias in responses.

2. Background and related work

2.1. Health chatbots

Health chatbots are AI powered conversational agents designed to assist patients, clinicians, or administrators through dialogue. Their interfaces are typically text or voice based, with recent models also being embedded in apps, web portals, and telehealth platforms. They support a broad range of applications in healthcare, such as patient triage, chronic disease monitoring, medication adherence, mental health support, and health education. The appeal of chatbots lies in their scalability and availability: they offer 24/7 assistance, reduce the burden on clinical staff, and provide timely responses to routine inquiries. In settings with limited resources, they can act as support systems, helping guide patients toward

appropriate care. During public health crises like the COVID-19 pandemic for instance, chatbots were used for symptom screening, vaccination scheduling, and information dissemination (Espinoza, Crown and Kulkarni, 2020).

Those chatbots systems were originally largely rule-based. This means they were based on predefined scripts and decision trees to simulate conversations, giving deterministic responses based on keyword matches or hierarchical question flows (Shawar and Atwell, 2007). They are simple to implement and useful for basic healthcare usages, but offer limited contextual understanding and thus cannot meet the demands of clinical communication (Radziwill and Benton, 2017).

With machine learning, chatbots are now more based on deep learning models and thus have a more dynamic, data driven language generation. A foundational work was the development of BERT (Bidirectional Encoder Representations from Transformers) by Devlin et al. (2018), which introduced a transformer architecture capable of learning context from both the left and right sides of a word in a sentence. This architecture really improved performance on natural language understanding tasks such as question answering, sentiment analysis, and named entity recognition.

Based on BERT's success, more specific models like BioBERT (Lee et al., 2020), DistilBERT (Sanh et al., 2019) and ClinicalBERT (Alsentzer et al., 2019) were implemented to address the unique vocabulary and structure of biomedical language. BioBERT is more effective for biomedical named entity recognition (NER), relation extraction, and medical question answering; and DistilBERT is a compressed version of BERT, more suited for low resource real time environments.

Despite these advancements, large language models have limitations. An important concern is the tendency to generate confident but incorrect outputs (Ji et al., 2023). Moreover responses can be sensitive to prompt phrasing, user demographics, or linguistic bias (Dušek, Novikova and Rieser, 2020). In high stakes fields like healthcare, this can have serious consequences. Misinterpretations can lead to diagnostic errors, misinformation, or user disengagement, especially with vulnerable populations such as the elderly, non-native speakers, or individuals with low health literacy.

These issues raise important questions about reliability, safety, and ethical deployment in clinical environments (Young et al., 2018; Mehrabi et al., 2021). These limitations make it hard to use AI in the full range of human diagnostic reasoning, and motivate the shift to multimodal models.

2.2. Multimodal models in medicine

Physicians use different inputs for diagnosis: patient interviews, imaging studies, lab results, physical examinations. In a similar way, multimodal machine learning builds models capable of combining different data types such as text, images, audio, and structured signals into one system.

In medicine, multimodal models have recently gained traction for tasks that need visual and textual data, most notably in radiology and dermatology. For example, BioViL (Boecking et al., 2022) is a vision-language model trained to align chest X-rays with corresponding radiology reports, and Med-PaLM M (Singhal et al., 2023) extends Google's Med-PaLM framework by incorporating image inputs. Another important development is LLaVA-Med (Zhang et al., 2023), which combines vision encoders with large language models to perform visual question answering, explain medical images, and interact with clinicians in a conversational form.

But multimodal integration also has its difficulties. The fusion of representations from different modalities requires alignment in terms of semantics, scale, and signal structure. There thus exist techniques like early fusion (combining inputs at the feature level), late fusion (combining model outputs), and cross-modal attention mechanisms (Zadeh et al., 2017; Kiela and Bottou, 2014). Additionally as models grow in complexity, interpretability and traceability become harder. And accountability and transparency are a key concern in clinical environments (Mitchell et al., 2021).

2.3. Bias in clinical AI

As AI systems increasingly assist healthcare practices, bias becomes an important subject to tackle. Bias in machine learning refers to systematic errors that unfairly disadvantage certain individuals or groups. In clinical AI, it can manifest in multiple ways, such as underdiagnosis, overtreatment, and exclusion.

Bias in the medical field can be conceptualized with three directions: data driven, algorithmic, and human. Data driven bias comes from unrepresentative or incomplete datasets. Algorithmic bias happens when model design, feature selection, or optimization objectives encode or amplify disparities. Finally, human bias is the influence of clinician judgment, healthcare infrastructure, and larger societal inequities on both data generation and the interpretation of outputs. Recognizing these interconnected sources of bias is essential for developing assessment and mitigation strategies that address not only technical limitations but also structural inequities within healthcare systems (Norori et al., 2021). Moreover, it is important to keep in mind that bias can occur at every stage of the machine learning pipeline (Barocas, Hardt, and Narayanan, 2019), from data collection and labeling to model training, evaluation and real-world deployment.

Another challenge is having to develop models with small datasets. This happens often in medical applications because of the rarity of certain conditions, privacy restrictions, or the high cost of expert annotation. By failing to capture the variability within patient populations, models generalize poorly to underrepresented groups. Indeed, small data scenarios increase the risk of overfitting, where models memorize noise in the training set rather than learning generalizable patterns (Shaikhina and Khovanova, 2017). This can disproportionately affect minority subgroups if their representation is sparse, ultimately reinforcing pre-existing healthcare bias.

In light of these challenges, our study seeks to contribute to the growing conversation on fairness in clinical AI by evaluating how unimodal and multimodal diagnostic models perform across controlled variations in patient demographics. We aim to analyze not only model performance, but also whether multimodal integration mitigates or magnifies biases. Given our available resources, we will focus on data driven and algorithmic sources of bias, looking into the distribution of errors across subgroups.

3. Methodology

This section outlines the data preprocessing, model architectures, and evaluation pipeline setups used to compare unimodal and multimodal diagnostic chatbots. We also describe the framework for assessing demographic fairness across models.

3.1. Data description and relevance

The dataset chosen is the Skin Lesion Dataset for Classification (Ribeiro et al., 2022), hosted on Mendeley Data. It was chosen because it includes both structured clinical metadata and corresponding dermoscopic images for over 2,000 patients, which is aligned with our objectives.

The data was collected from a Brazilian hospital system, and has diverse patient records that span age, gender, ethnicity, and socioeconomic status. Each entry includes binary indicators of lesion symptoms (e.g., pain, itch, bleeding), demographic information (e.g., age, gender, parents’ origin), some cancer history, and environmental factors (e.g., access to sewage infrastructure, to piped water). These variables are suited for bias evaluation, allowing us to analyze model behavior across population subgroups that are often underrepresented in AI datasets (Obermeyer et al., 2019). Every record has a diagnostic label representing the ground truth classification of the lesion, obtained through histopathological verification. This ensures that the ground truth reflects outcomes that are clinically validated.

Moreover, skin lesion classification is a clinically significant task in medical AI research. Dermatological diagnostics are often visual and rely a lot on a combination of patient history and image interpretation. So skin lesions are a well suited topic for our study. The field has also seen growing interest in automated triage tools and AI-assisted dermatology, where chatbots and clinical decision systems are used to improve early detection and healthcare access.

3.2. Data Preprocessing

To prepare the data, we first remove any record with a missing diagnostic label or invalid image reference. The six diagnostic categories are: actinic keratosis (ACK), basal cell carcinoma of skin (BCC), squamous cell carcinoma (SCC), seborrheic keratosis (SEK), malignant melanoma (MEL), and melanocytic nevus of skin (NEV). We map them to integer IDs for supervised learning. Because several metadata fields are stored as mixed types (0/1, True/False, or “yes/no”), we normalize all binary indicators.

For the textual input, an important step is that we convert each patient’s structured fields into a natural language prompt that includes all available clinical context. Missing fields are skipped so the prompt remains fluent without placeholders. This allows us to simulate a realistic conversational input that might be given to a diagnostic chatbot.

For the visual input, all non-existent or unreadable files are dropped. The remaining images are loaded as RGB and preprocessed with the visual encoder’s native normalization and resizing. For training only we apply light, dermoscopy appropriate augmentations such as horizontal flip or mild affine and brightness/contrast jitter to help models generalize.

3.3. Unimodal model

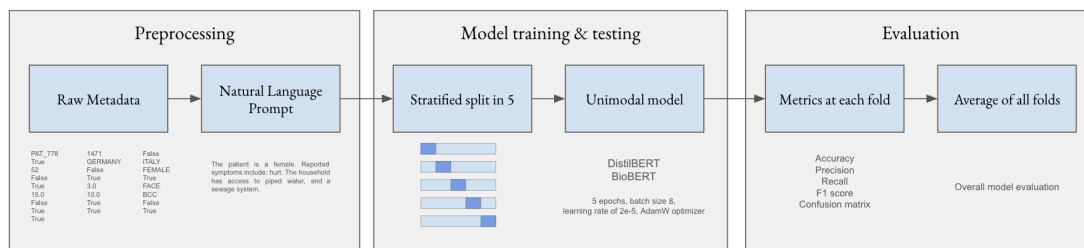


Figure 1.1: Pipeline for the unimodal diagnostic models

We first train text only models to quantify how much diagnostic signal is present in the structured metadata. Each patient’s prompt is tokenized with the model’s native tokenizer (maximum length 256 with truncation and padding). To assess the models, we report accuracy as well as macro and per-class F1, with the corresponding confusion matrices to understand systematic confusions.

As a general domain baseline we fine tune DistilBERT (distilbert-base-uncased). DistilBERT is a compact transformer distilled from BERT that retains most of BERT’s language understanding while using fewer parameters and running faster. Using DistilBERT also helps isolate the value of domain specific pretraining, allowing us to attribute gains to biomedical knowledge rather than sheer model size or capacity.

To represent a domain aware unimodal chatbot, we fine tune BioBERT (dmis-lab/biobert-base-cased-v1.1). It is a transformer based model pretrained on large scale biomedical corpora and is designed to handle clinical terminology and abbreviations more effectively than general domain encoders. BioBERT is made for strong performance on clinical NLP tasks, which makes it suitable for this scenario.

For both models, instead of a single train-test split, we use a 5 fold stratified cross validation approach. This is done to increase the reliability of our evaluation metrics, especially given the limited size and class imbalance present in the dataset. Stratification ensures that all diagnostic categories are proportionally represented in each fold, reducing variance across splits and minimizing the impact of outliers or rare conditions. For each fold, a new model is initialized and trained using the HuggingFace Trainer API. The training process runs for five epochs with a batch size of 8 and a learning rate of 2e-5. We use the AdamW optimizer with weight decay to prevent overfitting. For each fold, we evaluate using classification metrics: accuracy, precision, recall, and F1 score. We also generate confusion matrices and per-class reports to analyze diagnostic performance across different conditions. By aggregating results over all five folds, we get a more comprehensive view of the model’s generalization capability and stability.

3.4. Multimodal model

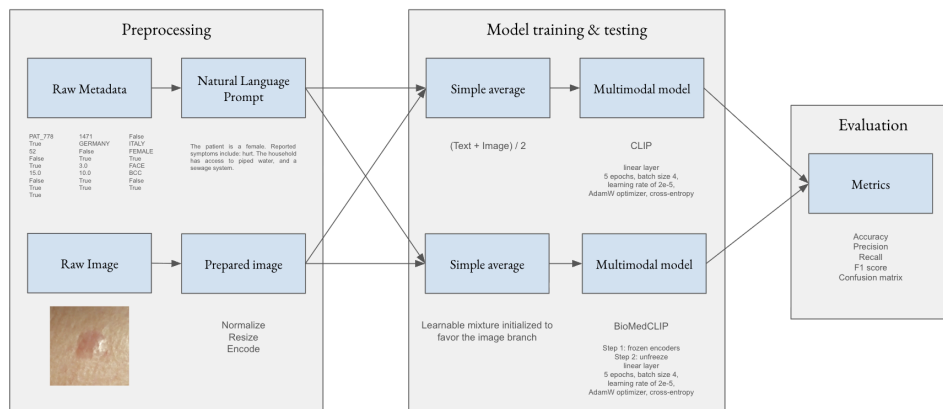


Figure 1.2: Pipeline for the multimodal diagnostic models

To evaluate the potential of multimodal diagnostic reasoning, we train two text+image models that combine each patient’s clinical prompt with the corresponding dermoscopic image.

As a general multimodal baseline, we fine-tune CLIP ViT-B/32 (the LAION variant) using Hugging Face’s CLIPProcessor/CLIPModel. CLIP is a dual encoder model trained to align visual and textual representations in a shared embedding space. Because CLIP expects short text, we keep each prompt within its standard 77 token limit and resize images to the model’s default size. We take the model’s image and text features, combine them with a simple average, and add a small linear layer to predict one of the six diagnoses. This modification allows the model to make discrete diagnostic predictions instead of getting similarity scores. Training is end to end with AdamW and cross-entropy, and we track the same metrics as for the unimodal models. This again gives us a solid reference system without any biomedical pretraining.

To better align the model with clinical language and medical imagery, we then fine tune BioMedCLIP (microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224) via open_clip. Compared to generic CLIP, BioMedCLIP’s PubMedBERT text tower supports a longer 256 token context, and its image+text pretraining is biomedical. We keep the same preprocessing but modify fusion. Indeed in addition to the fixed average, we use a learnable mixture initialized to favor the image branch, allowing the model to adaptively weight modalities. For stability, training happens in two steps: a short warm up phase where the encoders are frozen and so only the small classifier layer learns, followed by an unfreeze phase where the whole model learns with a small learning rate on the backbone and a larger one on the classifier. We also use a few standard safeguards: label smoothing, early stopping, and gradient clipping. Because the classes are imbalanced, we apply class weighted cross entropy. Like for the CLIP baseline, we pick the checkpoint with the lowest validation loss and report accuracy and macro/per-class F1, along with confusion matrices.

4. Results

This section evaluates our chosen models' diagnostic performance and the distribution of errors across demographic subgroups. All results are reported on held out data, with checkpoints selected by the lowest validation loss. Diagnostic performance is summarised using overall accuracy and macro/per-class F1, and bias is assessed by computing subgroup specific accuracies together with their sample sizes.

4.1. Diagnosis accuracy

The aim of this study pertains first is the diagnosis accuracy analysis of our models, i.e. if they diagnose skin lesions correctly based on the data they're provided. This is a crucial and high responsibility aspect of the consideration for implementation of AI chatbots in the healthcare sector.

Model	Overall test accuracy
DistilBERT	0.6575
BioBERT	0.6867
CLIPModel	0.7434
BioMedCLIP	0.7696

Figure 2: Test accuracy for all models

We can see in figure 2 that BioBERT outperforms DistilBERT within unimodal models, and BioMedCLIP outperforms CLIP within multimodal models. This highlights how domain specific training provides for better results. Moreover, multimodal models outperform unimodal models, which is the key aspect we wanted to verify in this study. And overall, BioMedCLIP has the best diagnosis accuracy amongst the models we selected for our study.

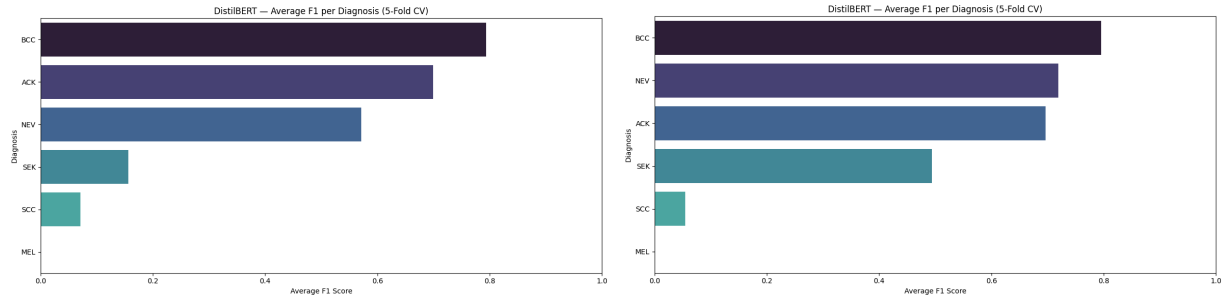


Figure 3.1: F1-scores for DistilBERT and BioBERT

Diagnosis	DistilBERT	BioBERT
BCC (845)	0.793438	0.795637
ACK (730)	0.699323	0.696585
NEV (244)	0.571208	0.719329
SEK (235)	0.156369	0.493907
SCC (192)	0.070634	0.054125
MEL (52)	0.000000	0.000000

Figure 3.2: F1-scores for DistilBERT and BioBERT

Figure 3.1 shows bar charts of the two unimodal models used, containing F1-scores for each skin lesion in our dataset. Figure 3.2 shows the details of said scores as well as the sample size of each skin lesion. So looking further into the diagnosis accuracy, we can see that DistilBERT and BioBERT have a similar F1 score for each skin lesion, with BioBERT being significantly better for NEV and SEK. Very notably, we observe that the bigger the sample size the better the F1 score for almost all lesions (exception being for BioBERT’s diagnosis of ACK). This shows that both models struggle to detect patterns in data for rarer lesions, both indeed failing to diagnose any cases of MEL.

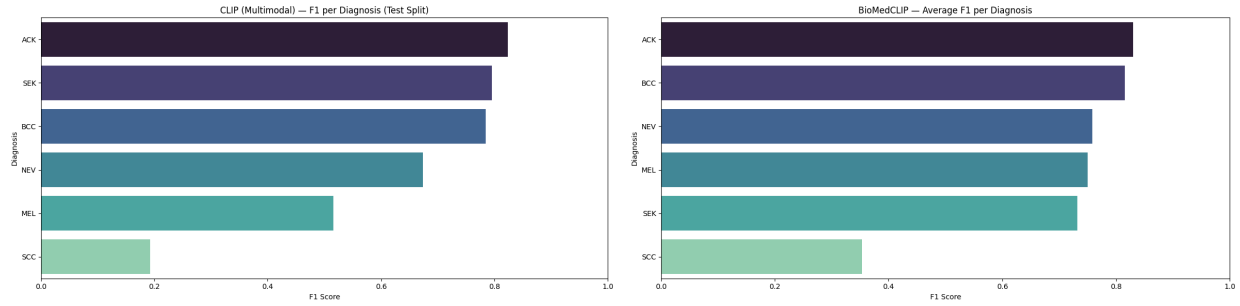


Figure 4.1: F1-scores for CLIPModel and BioMedCLIP

Diagnosis	CLIPModel	BioMedCLIP
BCC (845)	0.784615	0.815864
ACK (730)	0.823529	0.830325
NEV (244)	0.673913	0.758621
SEK (235)	0.795181	0.731707
SCC (192)	0.192308	0.352941
MEL (52)	0.516129	0.750000

Figure 4.2: F1-scores for CLIPModel and BioMedCLIP

Figure 4.1 and 4.2 show the same as above, but for the two multimodal models instead. We can see here that both multimodal models severely outperform the unimodal ones for all lesions. We also notice that here, the sample size has less impact on the F1 score, with for example SCC having by far the lowest scores. Our guess is that in the case of image analysis, lesions with more visible particularities are easier to learn on.

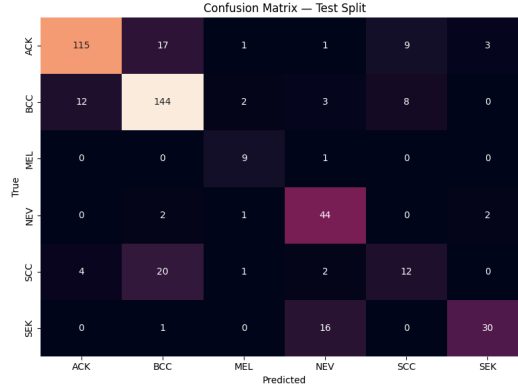


Figure 5: Confusion matrix for BioMedCLIP

And indeed, if we look at the confusion matrix, we notice that SCC tends to be confused for other lesions.

Overall, multimodal models have superior diagnosis power over skin lesions. Their accuracy is satisfying for most lesions, and there are none that stay totally undiagnosed. And so even if the results are far from perfect, this is of major importance as it shows extremely promising results in the capability of AI to assist real life physicians in their tasks.

4.2. Bias assessment

To evaluate fairness in diagnostic performance, we conduct an analysis of model predictions across key demographic variables. For this we take the two best models for each category (unimodal and multimodal), namely BioBERT and BioMedCLIP. For both models, we generate predictions for the entire dataset and compare them to the ground-truth diagnostic labels. We compute a binary correctness indicator for each prediction, allowing us to aggregate accuracy within subgroups.

We examine three main subgroup axes: gender, sewage system access as a proxy for socioeconomic status and parental origin as representation of ethnic or cultural backgrounds.

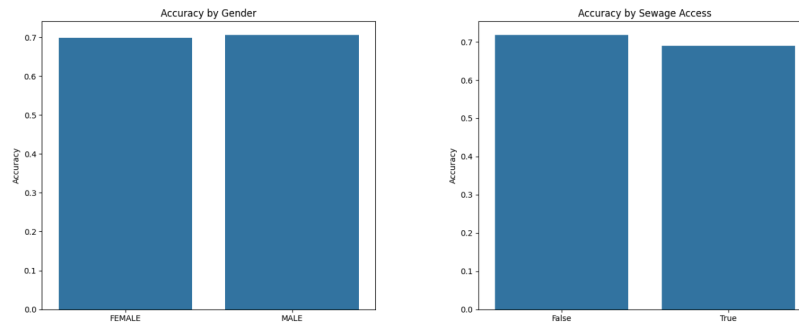


Figure 6.1: Accuracy by Gender and Sewage Access for BioBERT

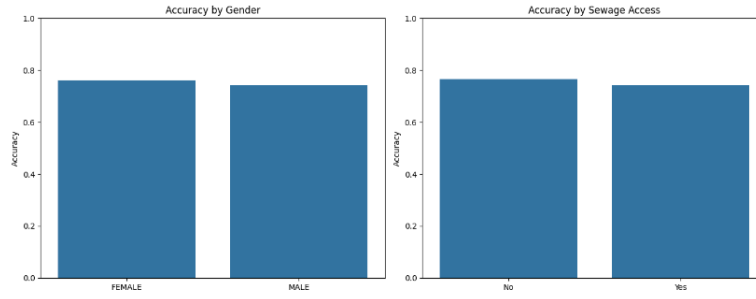


Figure 6.2: Accuracy by Gender and Sewage Access for BioMedCLIP

	BioBERT		BioMedCLIP	
Accuracy by Gender	Unknown (804)	0.7575	Unknown (804)	0.9453
	Male (741)	0.7058	Male (741)	0.7610
	Female (753)	0.6985	Female (753)	0.7422
Accuracy by Sewage Access	Unknown (804)	0.7575	Unknown (804)	0.9453
	No Access (648)	0.7176	No Access (648)	0.7654
	Access (846)	0.6903	Access (846)	0.7411

Figure 6.3: Detailed Accuracy by Gender and Sewage Access for BioBERT and BioMedCLIP

Figures 6.1, 6.2 and 6.3 show the accuracy by gender and sewage access for both models, with figure 6.3 being more detailed and also providing results for unknown genre and sewage access as well as overall sample size indications for more in depth analysis. As stated in the previous section, what we observe first is that the multimodal approach outperforms the unimodal one, and that for all gender and sewage access categories. Another important observation is that accuracy for both unknown categories skyrockets with the multimodal model. After review of those entries in the dataset, it becomes evident that this is not due to gender or sewage access but rather to the simplicity of those entries making their lesion easier to diagnose. Indeed they have less conflicting factors. As to bias analysis, even though we observe that male and female patients as well as patients with and without sewage access do not have the exact same accuracy results, these differences remain not significant enough to assess the presence of gender or (by proxy) socio-economic bias in this study. Furthermore, the gap comparison between unimodal and multimodal model accuracy is also too little to be conclusive with respect to bias assessment.

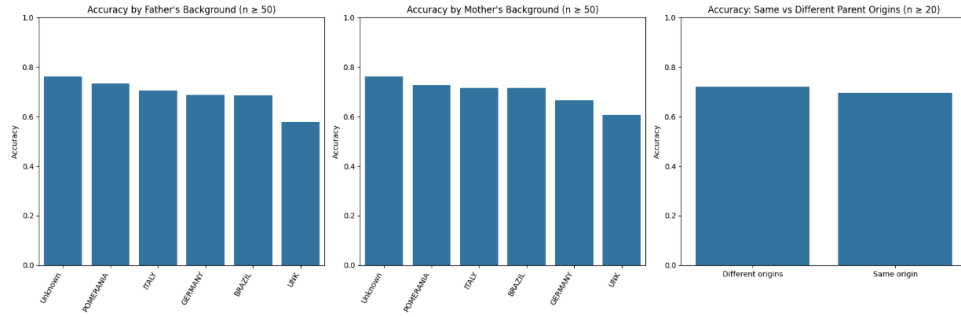


Figure 7.1: Accuracy by Parental Origin for BioBERT

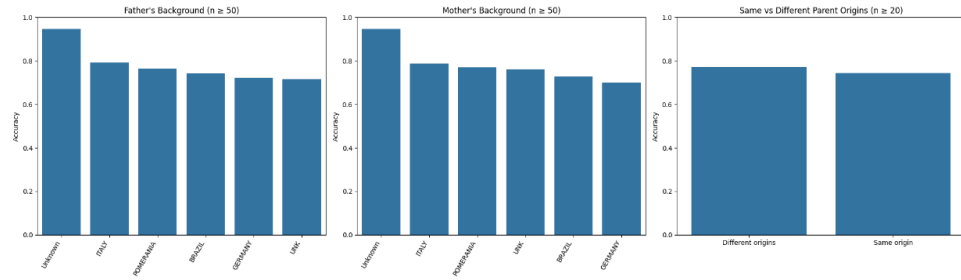


Figure 7.2: Accuracy by Parental Origin for BioMedCLIP

	BioBERT		BioMedCLIP	
Accuracy by Father's Background (n>=50)	Unknown (913)	0.7415	Unknown (913)	0.9222
	Pomerania (530)	0.7321	Italy (250)	0.7920
	Italy (250)	0.7040	Pomerania (530)	0.7642
	Germany (466)	0.6867	Brazil (89)	0.7416
	Brazil (89)	0.6854	Germany (466)	0.7232
Accuracy by Mother's Background (n>=50)	Unknown (906)	0.7461	Unknown (906)	0.9294
	Pomerania (533)	0.7261	Italy (236)	0.7881
	Italy (236)	0.7161	Pomerania (533)	0.7692
	Brazil (81)	0.7160	Brazil (81)	0.7284
	Germany (482)	0.6660	Germany (482)	0.7012
Accuracy by Combined Parent Origin (n>=20)	Different (236)	0.7203	Different (236)	0.7712
	Same (1238)	0.6955	Same (1238)	0.7439

Figure 7.3: Detailed Accuracy by Parental Origin for BioMedCLIP

Figures 7.1, 7.2 and 7.3 show the accuracy by parental origin for both models, with figure 7.3 being more detailed and also providing results for unknown origins as well as overall sample size indications for more in depth analysis. For relevance, we only select data points starting from a minimum sample size. Again we observe that the multimodal model consistently does better than the unimodal one, and that unknown origin data points have better accuracy (for the same reason as stated above). Contrary to before, we now see that there is a slightly noticeable gap in diagnosis accuracy depending on parental origin, with Pomeranians and Italians being consistently better diagnosed than Germans and Brazilians by around 5 percentage points (difference between best diagnosed and worst diagnosed known origin). This gap pertains for both models. And even though the difference is not extremely significant, it is enough to conclude to the possibility of bias. We also identify a very slight gap (2-3 percentage points) within accuracy of whether both parents have the same or a different origin.

5. Discussion

5.1. Main findings and relation with literature

In our experiments, text+image models consistently outperform text only models, with BioMedCLIP achieving the highest test accuracy and F1 scores. This pattern aligns with prior work on multimodal learning, showing that combining complementary modalities improves performance on complex tasks (Ngiam et al., 2011; Baltrušaitis, Ahuja & Morency, 2019; Liang et al., 2022). The superiority in performance of domain specific tools seen through baseline comparisons with general ones underlines that they are to be developed and used for better performance in medical tasks. This result was expected but is nonetheless an agreeable intuition confirmation as well as is in accord with literature evidence that biomedical pretraining improves clinical NLP (Lee et al., 2020; Alsentzer et al., 2019).

Another finding is that unimodal models underperform on rarer or more confusable skin lesions, and that with addition of images this dependence on small sample size weakens but does not vanish. This is consistent with the small data generalization issues reported in medical machine learning studies (Shaikhina & Khovanova, 2017) as well as with reports that multimodal pretraining improves but does not fully resolve medical distinctions (Boecking et al., 2022; Singhal et al., 2023; Zhang et al., 2023).

During bias assessment, the better results obtained for unknown genre, sewage access, and origin lead us to believe that they are not factors that necessarily contribute a lot to healthcare bias in this study. Thus even if bias empirically does exist within these rough categories in the field, this study and the small gaps observed in the origin bias study indicate that the bias in this particular skin lesion diagnosis is neither algorithmic nor data driven. As the models are not designed to amplify disparities and the dataset is equally complete independently of genre, sewage access or origin, it is likely that the bias comes from a more intricate human side: with these AI chatbot systems there would not be significant difference in patient result at this particular step of the health treatment based on our three bias criteria. We can identify but we do lack the information to claim definitive conclusions as to the origin of the bias. This matches bias frameworks that differentiate data driven and algorithmic sources from human/structural ones and write that missing data can skew subgroup comparisons (Barocas, Hardt & Narayanan, 2019; Norori et al., 2021; Mehrabi et al., 2021).

5.2. Limitations

This study has several important limitations that should be acknowledged both in terms of performance and bias assessment.

First, the dataset used for training and evaluation is relatively small, which limits the models' capacity to generalize and increases the risk of overfitting.

Second, the dataset has an imbalanced class distribution, with some diagnostic categories and demographic subgroups being over and under represented. These disparities can and are affecting the learning process. As a result, the models perform better on well represented lesions and struggle with rarer ones.

Third, the overall predictive accuracy of our models remains far from perfect. This can limit the sensitivity of our bias detection. When misclassifications happen frequently across all groups, it becomes more difficult to know whether poorer performance in a subgroup is due to bias or simply the model's general unreliability.

Finally, our bias assessment methodology relies on relatively simple evaluation criteria: three group wise accuracy comparisons. While this provides an accessible first step in identifying disparities, more advanced fairness metrics and causal analyses could uncover deeper and more nuanced patterns of bias that remain hidden under our current approach.

5.3. Future research

To build from what we have tested and analyzed in this paper, as well as to remedy some of the limitations addressed, future research should in priority extend to larger and more balanced multimodal datasets. A broader dataset would improve predictive performance and enable a more rigorous bias assessment. Another important improvement would be to investigate bias factors more effectively, both during dataset selection and during evaluation. With richer socio-demographic indicators such as geographic location, postal codes, or socioeconomic markers, we could find more valuable insights into structural determinants of health.

On the technical side, exploring more sophisticated multimodal architectures could also have its benefits. We could perhaps achieve better diagnostic accuracy with different models or fine tuning strategies, or more domain adaptive pretraining on dermatology specific corpora. We could also make use of clinical validation through active learning and the inputs from real life clinicians.

Beyond accuracy, another point of research would be to study how these systems can be implemented in the field to help actual patients and clinicians. This could go through testing in realistic triage settings to check for speed, cost-effectiveness and usefulness, with possible recalibration from human expertise. It also means designing accessible, adapted and modular interfaces. And taking into account the scale and importance of the field, a lot of work is to be explored in terms of privacy, anonymity and user consent.

Together, these directions would provide a stronger foundation for the development of multimodal diagnostic chatbots that are accurate, fair, clinically relevant and socially beneficial. All in all, there's a lot of promising and exciting work to be done for these systems to become an industry-enhancing tool.

6. Conclusion

This study aimed to determine whether multimodal AI chatbots improve skin lesion diagnosis over text only models and whether multimodality helps mitigate demographic bias across patient subgroups. Our results show that multimodal and domain specific models perform better than unimodal ones, while observed subgroup gaps appear more linked to data scarcity and intricate bias than to algorithmic amplification. These findings contribute to the ongoing development of AI in healthcare by supporting multimodal, clinically adapted tools and by outlining practical fairness checks for deployment.

7. References

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